

HAZEM KHALED EZAT

Applied AI & Data Analyst · Python · SQL · Power BI · Deep Learning · Front-End Development

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SUMMARY

I'm Hazem — a data analyst who taught himself frontend development, then went deeper into data and AI through the Applied AI & Data Analytics Diploma (Digital Pioneers / MCIT). Right now I build KPI dashboards, SQL pipelines, predictive models, and deep learning systems. I led teams on all 4 course projects and my graduation project — DigiSteel-YOLO, a deep learning model for steel defect detection. I'm also deep into the AI tools ecosystem: I build autonomous agents, design MCP integrations, and create reusable agent skills. I lead naturally, I communicate well across technical and non-technical people, and I build things other people can actually pick up and run.

WORK EXPERIENCE

Data Analyst

Digilians | MCIT · Cairo, Egypt · Dec 2025 – Aug 2026

- Built KPI dashboards in Power BI that made recurring metrics self-serve, reducing the volume of one-off reporting requests.
- Developed SQL reporting workflows (PostgreSQL) that replaced manual data preparation steps and improved turnaround time on recurring reports.
- Designed relational schemas with full ERD documentation, improving consistency and traceability of data used in downstream reporting.
- Presented findings to non-technical stakeholders through custom visuals designed to be readable without additional explanation.

SELECTED PROJECTS

1. DigiSteel-YOLO — Deep Learning for Industrial Defect Detection

Python · PyTorch · Ultralytics YOLO · Computer Vision

github.com/hazemelerefey/DigiSteel-YOLO

Built a custom YOLO-based deep learning model for detecting defects on steel surfaces using the NEU-DET dataset. Designed a robustness testing framework with 24 test scenarios (6 perturbation types, 4 severity levels) — something other steel defect papers haven't done. The project includes modular training pipelines, attention mechanisms, and configurable augmentation.

2. Churn Prediction Model

Python · Scikit-learn · SHAP · Pandas

github.com/hazemelerefey/analytics-churn-prediction

Churn prediction model using Scikit-learn, with SHAP-based interpretation to explain which factors drive customer churn. Covered the full pipeline: cleaning, feature prep, training, and evaluation.

3. Global E-Commerce Sales Tracker

Power BI · SQL · Data Modeling

github.com/hazemelerefey/global-ecommerce-sales-tracker

Power BI dashboard for tracking revenue, profit, orders, and regional performance. Built for executives who need KPIs at a glance, with drill-downs for deeper analysis.

4. Social Intelligence Pipeline — Autonomous AI Agent

n8n · REST APIs · RSS/Atom Parsing · Reddit API · Cron Automation

github.com/hazemelerefey/n8n-projects

An autonomous agent that runs every morning, pulls trending posts from 5 subreddits (technology, startups, SaaS, webdev, productivity), handles rate limits and retries on its own, and normalizes everything into a clean format. Built to run without supervision.

5. Social Content AI Agent — Multi-Platform Publishing

n8n · OpenAI API · Hacker News API · Dev.to API · Facebook · X (Twitter)

github.com/hazemelerefey/n8n-projects

An autonomous agent that finds trending stories from Hacker News and Dev.to, scores them with a custom relevance algorithm, then generates Arabic content for Facebook and X. It has its own brand voice, a 6-angle content strategy, and publishes twice a day at 10 AM and 6 PM — no manual work needed.

6. Personal Portfolio Website — Harry Potter Edition

Next.js · TypeScript · CSS · Vercel

hazemelerefy.vercel.app · Source

My personal portfolio, built with Next.js and TypeScript, themed after Harry Potter. Custom SVGs, animations, responsive layout. Deployed on Vercel.

SKILLS

Analytics: SQL (PostgreSQL, SSMS) · Python (Pandas, NumPy, Scikit-learn) · Power BI · Excel (Advanced) · DAX · ETL · Star Schema · Data Warehousing · Statistical Analysis

Machine Learning & AI: Deep Learning (PyTorch, YOLO) · Classification Models · SHAP Analysis · Feature Engineering · Predictive Modeling · Model Evaluation (AUC, F1, Confusion Matrix)

Applied AI & Automation: Autonomous AI Agents · n8n Workflow Automation · MCP Server Design & Integration · Agent Skill Development · Plugin Architecture · MLOps Fundamentals · End-to-End ML Pipelines

AI Tools (Expert): Claude Code CLI · OpenClaw · Hermes CLI · Antigravity · Qoder · OpenCode · Mimo Code · Cursor · Windsurf · Copilot

Front-End: HTML5 · CSS3 (SASS, Tailwind) · JavaScript (ES6+) · React.js · Next.js · Figma · REST API Integration

Tools: Git · GitHub · Jupyter Notebook · Google Colab · VS Code · Streamlit

Soft Skills: Leadership · Leadership Without Authority · Cross-Functional Communication · Strategic Thinking · Team Collaboration · Problem-Solving · Self-Directed Learning

LANGUAGES

Arabic: native · English: Intermediate (B1)

EDUCATION

Diploma, Applied AI & Data Analytics

Digilians | MCIT · Dec 2025 – Aug 2026

Python, SQL, Machine Learning, Deep Learning, and AI Agent Systems.

Nanodegree, Front-End Web Development

Egypt FWD (Udacity) · Sep 2020 – Mar 2022

Bachelor of Laws (LL.B.) — Commercial & Corporate Law

Port Said University · 2020 – 2024

Business regulations, corporate governance, and commercial contracts. Legal training provided structured reasoning, research methodology, and regulatory compliance foundations that strengthen work in data governance, business rules, and analytics integrity.

CERTIFICATIONS

Delivering Quality Work with Agility

IBM (Coursera)

Google Data Analytics Professional Certificate

Google (Coursera)

Foundations: Data, Data, Everywhere · Ask Questions to Make Data-Driven Decisions · Prepare Data for Exploration